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Assignment 2

Research Objective & Motivation

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1. Background & Context

Sentiment Analysis (or opinion mining) is a crucial application of Natural Language Processing (NLP) that aims to detect, retrieve and categories of opinions, attitudes and emotional expressions from text-based information. It helps computers to understand language and identify the mood or attitude expressed in written text. Generally, sentiment analysis can categorize texts into three main types positive, negative, and neutral. Positive sentiment, positive opinions or feelings negative sentiment dissatisfied or negative opinions neutral sentiment a statement that does not reveal a clearly positive or negative opinion. More sophisticated sentiment analysis techniques may also be applied to detect specific emotions like happiness, sadness, anger, fear, and surprise. With NLP techniques, computers can handle and analyze vast quantities of text information that would otherwise take a long time to analyze.

Digital communication has expanded at a fast pace and social media, especially Facebook, Instagram, X (previously known as Twitter) and YouTube are major sources for people to post their opinions, experiences, reactions, and emotions. Users create a tremendous number of words each day, whether it's in comments, posts, reviews, or online discussions. These user-generated contents help shape information about public opinions and emotional reactions to a variety of topics (products, services, social issues, entertainment, and current topics). Thus, sentiment analysis of social media data became more relevant to comprehend and gain insights into people's attitudes and determine their sentiments on a massive scale by analyzing text data on social media. For instance, companies can review their customer feedback and product reviews to determine satisfaction or dissatisfaction, and organizations can look at the reactions of the public regarding campaigns, events, and social issues. Likewise, by analyzing online discussions, sentiment analysis can be used to identify the public's opinion and emotions about specific subjects.

Although useful, sentiment analysis of social media text is hindered by the informal and unstructured text format of social media communication. Social media users are not always careful to use formal written language they often use slang, abbreviations, misspelled words, repeated characters, informal language and combinations of languages in their communication. Such differences can make it challenging for NLP systems to accurately grasp a sentence's meaning and tone. Besides, the sentiment expressed in the social media produced content isn't always conveyed through words. Emojis, emoticons, punctuation, and

other non-textual elements are common features in the user's messages, which they use to convey their emotions or give their messages more emotional content. For example, a sentence that includes positive words can be sarcasm or dissatisfaction based on its context and its accompanying emoji. Therefore, using just the words used in a social media post may not be enough to correctly infer its sentiment.

Furthermore, the properties of social media data can affect the performance of sentiment classification. There could be an imbalance of sentiment classes, with some classes being much larger than others in social media datasets. This may impact on the performance of machine learning models for recognizing less frequent sentiments. The multilingual nature of social media presents significant challenges in understanding its nuances, complexity, and scale. This has led researchers to implement a wide range of approaches to social media NLP, including the use of traditional machine learning models like Support Vector Machines (SVM) and Logistic Regression, as well as more recent deep learning and Transformer-based models like BERT and RoBERTa. The methods enable computers to detect linguistic patterns, to acquire information about contexts, as well as to categorize text content based on its expressed sentiment. Hence, the sentiment analysis by NLP becomes an important tool to determine the automatic emotional or opinion analysis of social media news texts.

2. Problem Statement

Despite advances in transformer-based architectures and deep learning methods, social media sentiment analysis models are still highly vulnerable. Based on our systematic literature review we show that there are three critical factors that limit the reliability of sentiment classification.

2.1 Inconsistent Preprocessing and Systematic Stripping of Non-Textual Features:

Non-linguistic characters such as emojis, emoticons and repeated punctuation marks are commonly removed in preprocessing stages of traditional social media. However, emojis are often used as a primary source of emotion, sarcasm and sentiment by social media users. In fact, retaining emojis as a semantic unit improves accuracy significantly with an increase of up to 7.46% (85.98% accuracy with emojis compared to 78.52% without emojis) in Arabic microblogs (Dadure et al., 2025). Non-linguistic terms cause information loss during preprocessing stages resulting in misclassification of sentiment by classifiers due to the lack of sentiment dense information in very short texts.

2.2 Severe Class Imbalance and Majority-Class Prediction Collapse:

In real-life scenarios, we often encounter class imbalance problems. There are many more positive or neutral comments than negative or mixed sentiments. For example, in benchmark settings (Eusha et al., 2024), more than two-thirds of examples used for training were positive. As a result, deep neural networks and fine-tuned transformer models were over-predicting majority classes, and collapsing on the minority classes. This caused a very low Macro F1 score of 0.23 in the Tamil comments. If the model is trained using cross entropy loss (without any cost sensitive methods), high overall accuracy can be misleading. The model performs extremely poorly on crucial minority sentiments such as consumer complaints.

2.3 Confounding Experimental Factors in Existing Works:

While some works have accounted for the impact of emojis (Dadure et al., 2025) and class imbalance (Eusha et al., 2024 Hasan et al., 2023), they have been tested in different settings. The non-textual features and class imbalance were investigated in different languages, datasets, pre-processing methods and model complexities. Therefore, it is impossible to compare the individual and interaction effects of non-textual feature preservation and class weighting. It is unclear whether the improvements obtained in benchmarks are due to the complexity of model architecture or practical pre-processing and loss changes made at the data level.

3. Research Gap

Existing studies have revealed that the methods of preprocessing dataset can substantially affect the social media sentiment classification. However, the reviewed literature demonstrates inconsistency in terms of how sentiment-bearing non-textual information is handled. Some studies remove emojis, punctuation, and special characters as part of the text cleaning, while others retain emojis, convert them into textual representations, or use their sentiment polarity as explicit features. This inconsistency in preprocessing makes it difficult to identify whether removing such information improves classification by reducing noise or instead weakens performance by discarding useful emotional context. The findings reported by Dadure et al. (2025), where classification accuracy was higher when emojis were retained compared to when they were removed, further suggest that emoji information may have a meaningful influence on sentiment prediction. However, existing studies use different datasets, languages, models, and preprocessing strategies, hence this conclusion cannot be generalized.

A similar limitation also exists in the treatment of class imbalance. Several of the reviewed social media datasets contain significantly more samples from one sentiment class, specifically the positive class than other classes. Eusha et al. (2024) reported severe imbalance in their code-mixed sentiment datasets, while Hande et al. (2021) also identified strong positive-class dominance in the KanCMD dataset. This imbalance can cause classifiers to bias towards the majority class while performing poorly on minority classes such as negative, neutral, or mixed classes. Although previous studies pointed out class imbalance as

an important factor affecting classification performance, its effect is often observed together with other differences in language, dataset size, model architecture, and preprocessing. As a result, it remains difficult to conclude how much of the observed performance difference is specifically associated with class imbalance.

Most importantly, the reviewed studies do not provide a consistent basis for examining the relationship between these two factors. Emoji treatment and class distribution are often discussed or evaluated under different experimental conditions rather than being examined within the same controlled settings. Hence, when one study reports higher performance than another, the difference may result from the model architecture, dataset characteristics, language, preprocessing decisions, class distribution, or combination of these factors. This makes it difficult to isolate individual influence of retaining sentiment-bearing emoji information and handling imbalanced sentiment classes.

Thus, a practical research gap remains in the controlled examination of both factors under comparable experimental conditions. To be precise, existing evidence does not clearly prove whether retaining sentiment-bearing emoji information improves classification consistently, whether addressing class imbalance improves the recognition of minority sentiment classes, or whether both factors combined provides a more reliable sentiment classification than addressing each issue independently. By addressing this gap, it provides a clearer understanding on how preprocessing and class distribution could influence sentiment classification without the requirement to make changes in model architecture to explain the observed differences.

4. Research Objectives (ROs)

- RO1: To evaluate the impact of retaining versus stripping sentiment-bearing emoji features during preprocessing on social media sentiment classification performance under a fixed model architecture.

- RO2: To investigate the effectiveness of class-imbalance mitigation techniques in preventing majority-class prediction collapse and improving minority-class recognition across skewed sentiment distributions.

- RO3: To examine the combined interaction effect of emoji-preserving preprocessing and class-balanced training within a unified, controlled experimental framework without altering model architecture complexity.

In alignment with the identified research gap, these three objectives establish a step-by-step experimental design to isolate data-level and loss-level factors that are typically confounded in existing literature. RO1 directly addresses Problem 2.1 by holding the classifier, dataset, and training configuration constant while comparing text-only preprocessing where emojis and special characters are stripped, as in (Eusha et al., 2024) against emoji-preserving preprocessing where emojis are retained as semantic tokens, as in Dadure et al., 2025. RO2 addresses Problem 2.2 by evaluating how cost-sensitive learning or resampling strategies affect per-class recall and Macro-F1 on underrepresented negative, neutral, and mixed sentiment classes compared to standard cross-entropy training. Finally, RO3 addresses Problem 2.3 by testing both interventions jointly under identical experimental conditions, determining whether preserving non-textual emotional cues and balancing class weights yield synergistic gains in overall and minority-class sentiment prediction without requiring heavier model architectures.

5. Motivation & Expected Contributions

The primary motivation for this study stems from the growing disconnects between architectural complexity and practical reliability in social media sentiment analysis. Recent NLP studies increasingly rely on computationally expensive continual pretraining on hundreds of millions of posts (Barbieri et al., 2022) or complex multi-transformer ensembles (Hasan et al., 2023 Sharma et al., 2023) to achieve marginal benchmark gains. Despite this high computational cost, even advanced transformer models suffer severe performance degradation in real-world social media environments when fundamental data-level challenges are overlooked. Specifically, conventional text-cleaning pipelines routinely delete emojis and expressive punctuation that carry essential emotional and sarcastic cues in short texts (Eusha et al., 2024 Dadure et al., 2025), while naturally skewed class distributions cause classifiers to over-predict majority positive labels and fail on minority classes (Eusha et al., 2024). Prior studies evaluate these factors across disparate languages, datasets, and architectures, researchers and practitioners lack clear evidence on how much performance can be recovered purely through disciplined preprocessing and class-imbalance mitigation. This research is therefore motivated by the need to establish controlled empirical evidence that isolates the individual and combined value of non-textual feature retention and class-balanced learning.

5.1 Academic & Theoretical Contributions

First, this study contributes a controlled ablation framework that eliminates confounding experimental variables in social media sentiment benchmarking. In existing literature, performance differences between studies that strip emojis (Eusha et al., 2024) and those that retain them (Dadure et al., 2025 Barbieri et al., 2022) cannot be directly attributed to preprocessing alone because the underlying datasets, languages, sentiment granularities, and model architectures also differ. By evaluating emoji treatment and class-balancing strategies within a single standardized pipeline, this research isolates the precise variance explained by data representation versus class distribution.

Second, the study advances theoretical understanding of the interaction between paralinguistic features and minority-class learning. While Dadure et al. (2025) demonstrated that retaining emojis improves binary accuracy by 7.46% on balanced Arabic tweets, and Eusha et al. (2024) highlighted how class skewness degrades multi-class Macro-F1 to 0.23 in Tamil comments, the interplay between these two phenomena remains unexplored. This study provides empirical insights into whether sentiment-dense non-textual tokens such as negative or ambiguous emojis specifically aid classifiers in distinguishing hard-to-learn minority classes like negative, neutral, or mixed sentiments when paired with class-weighted training objectives.

Third, this work establishes evidence-based preprocessing guidelines for data-centric social media NLP. Rather than treating text normalization as a secondary technical detail, the findings clarify when non-textual characters function as informative semantic anchors versus noise, offering a reproducible methodological baseline for future multi-class emotion and sentiment studies (Camacho-Collados et al., 2022 Advani et al., 2020).

5.2 Practical & Industry Impact

From an industry perspective, the most critical contribution of this research is improving the detection of minority negative and mixed sentiments in real-world monitoring systems. On commercial social media platforms such as X, YouTube, Instagram, and Facebook, user-generated comments are often heavily skewed toward positive or casual reactions, leaving genuine customer complaints, product defects, and crisis signals as a small minority class (Eusha et al., 2024). A model that achieves high overall accuracy by defaulting to majority positive predictions provides false assurance to businesses. By combining class-imbalance mitigation with emoji retention, organizations can significantly improve recall on dissatisfied or frustrated user posts, enabling timely customer service intervention and accurate brand reputation management.

Furthermore, this research improves automated interpretation of informal, brief, and sarcastic user feedback. Everyday social media users frequently communicate dissatisfaction, irony, or mixed feelings by pairing short or seemingly positive phrases with contrasting emojis and punctuation rather than writing explicit opinion words (Camacho-Collados et al., 2022 Hasan et al., 2023). Demonstrating how to preserve and leverage these non-textual signals within standard NLP pipelines helps industry sentiment tools capture authentic user intent across product reviews, public policy discussions, and marketing campaigns.

Finally, the findings support cost-effective NLP deployment for resource-constrained organizations. Training and deploying massive multilingual transformers or multi-model ensembles demand substantial GPU memory and computational budgets (Barbieri et al., 2022). By proving that systematic data-level refinements specifically retaining emoji tokens and applying cost-sensitive class weighting can substantially boost Macro-F1 without increasing model size, this study offers a practical, low-cost pathway for businesses and developers to build reliable sentiment classifiers on standard hardware.

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